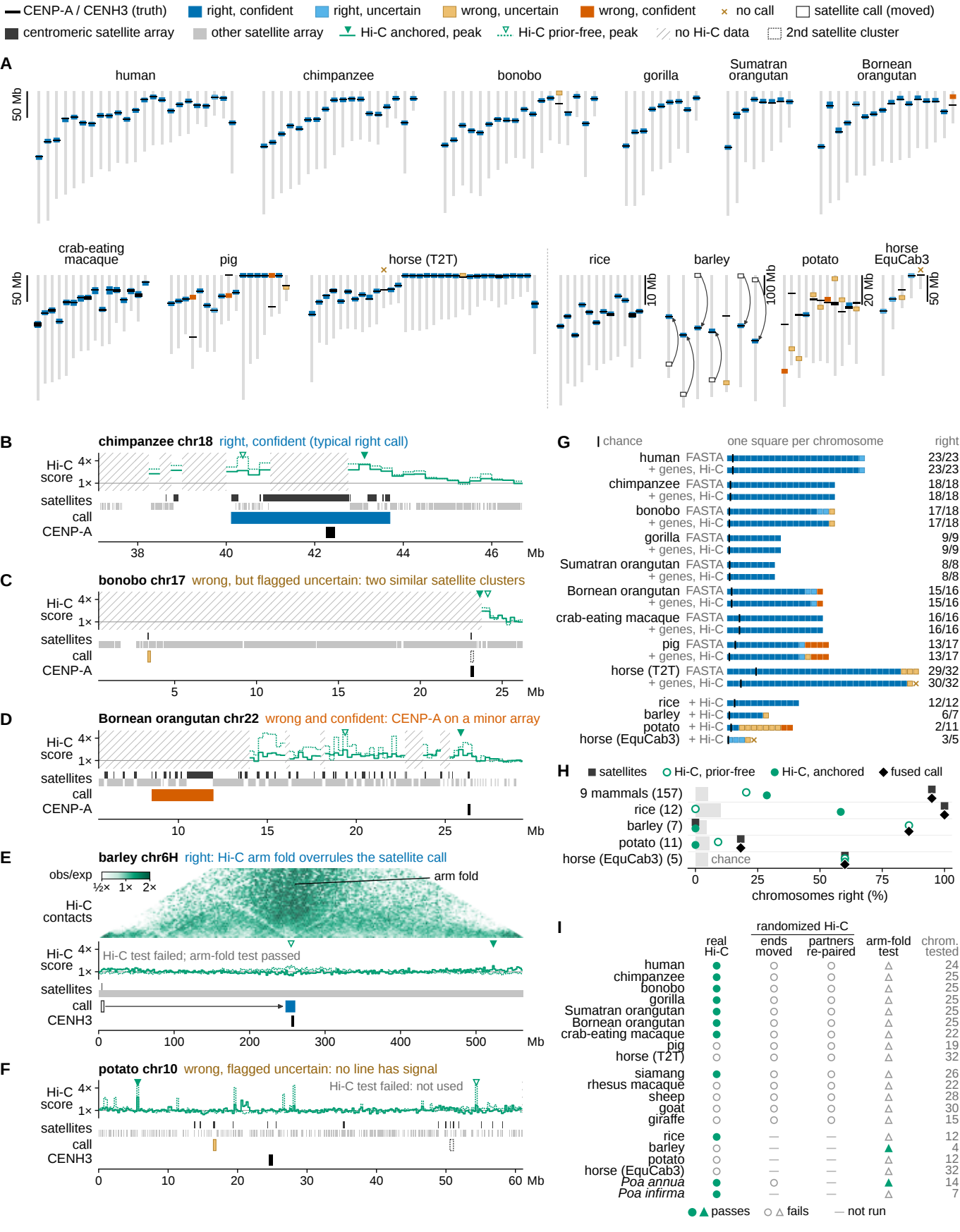

CENP-A / CENH3 (truth)
right, confident
right, uncertain
wrong, uncertain
wrong, confident
no call
satellite call (moved)
centromeric satellite array
other satellite array
Hi-C anchored, peak
Hi-C prior-free, peak
no Hi-C data
2nd satellite cluster
A
human
chimpanzee
bonobo
gorilla
Sumatran orangutan
Bornean orangutan
crab-eating macaque
pig
horse (T2T)
rice
barley
potato
horse EquCab3
50 Mb
10 Mb
100 Mb
20 Mb
B
chimpanzee chr18 right, confident (typical right call)
C
bonobo chr17 wrong, but flagged uncertain: two similar satellite clusters
D
Bornean orangutan chr22 wrong and confident: CENP-A on a minor array
E
barley chr6H right: Hi-C arm fold overrules the satellite call
obs/exp
arm fold
Hi-C contacts
Hi-C test failed; arm-fold test passed
F
potato chr10 wrong, flagged uncertain: no line has signal
Hi-C test failed: not used
Hi-C score
satellites
call
CENP-A
CENH3
Mb
G
chance
one square per chromosome
right
human FASTA 23/23
+ genes, Hi-C 23/23
chimpanzee FASTA 18/18
+ genes, Hi-C 18/18
bonobo FASTA 17/18
+ genes, Hi-C 17/18
gorilla FASTA 9/9
+ genes, Hi-C 9/9
Sumatran orangutan FASTA 8/8
+ genes, Hi-C 8/8
Bornean orangutan FASTA 15/16
+ genes, Hi-C 15/16
crab-eating macaque FASTA 16/16
+ genes, Hi-C 16/16
pig FASTA 13/17
+ genes, Hi-C 13/17
horse (T2T) FASTA 29/32
+ genes, Hi-C 30/32
rice + Hi-C 12/12
barley + Hi-C 6/7
potato + Hi-C 2/11
horse (EquCab3) + Hi-C 3/5
H
satellites
Hi-C, prior-free
Hi-C, anchored
fused call
9 mammals (157)
rice (12)
barley (7)
potato (11)
horse (EquCab3) (5)
chance
chromosomes right (%)
I
randomized Hi-C
real Hi-C
ends moved
partners re-paired
arm-fold test
chrom. tested
human 24
chimpanzee 25
bonobo 25
gorilla 25
Sumatran orangutan 25
Bornean orangutan 25
crab-eating macaque 22
pig 19
horse (T2T) 32
siamang 26
rhesus macaque 22
sheep 28
goat 30
giraffe 15
rice 12
barley 4
potato 12
horse (EquCab3) 32
Poa annua 14
Poa infirma 7
passes
fails
not run

Letting independent evidence lines check each other places centromeres correctly on nearly every satellite centromere, flags hard cases as uncertain instead of guessing, and fails openly where no line has signal; no single line works everywhere, and fusion keeps whichever line works.

A All 192 chromosomes with a mapped functional centromere, drawn to scale. Colored boxes are cenfuse's calls, black bars the measured CENP-A or CENH3 domain. In mammals and rice box and bar coincide; barley's calls move away from a misleading satellite; potato's centromeres are mostly missed, and flagged.

B–F Five chromosomes up close, one per kind of outcome: a typical right call on the centromeric satellite array (B); two similar satellite clusters, reported as uncertain (C); CENP-A on a minor array far from the main one, missed with confidence (D); a misleading satellite overruled by the folding of the chromosome arms seen in Hi-C (E); a potato centromere with nothing to follow (F).

G Every chromosome as one square. Nearly all mammal centromeres are placed right from the DNA sequence alone, far above chance; adding genes and Hi-C changes little beyond confidence.

H Each evidence line on its own. Satellites work in mammals and rice, Hi-C in barley, neither in potato; the fused call matches the best line in every genome.

I Hi-C is used only after it proves itself: its peaks on the other chromosomes must land on their own satellite calls more often than chance. It does in apes, crab-eating macaque, rice and both *Poa* species, and never on randomized Hi-C.

Terms. Truth: CENP-A domains (centromere dip regions in great apes; CENP-A ChIP-seq in crab-eating macaque, pig and horse T2T, for pig and horse from other animals) and CENH3 or CENP-A ChIP-seq domains (rice, barley, potato, horse EquCab3). Right: the call overlaps the truth (mammal satellite and final calls) or lies within four Hi-C bins of it (every plant call, and Hi-C peaks, which sit beside the unmappable array; bins 250 kb, human 500 kb, barley 2 Mb). Confident: cenfuse confidence high or medium; uncertain: low. Chance: the probability that a call of the same length placed at random on the chromosome is right, summed over chromosomes. Every call is shown and none is hidden by confidence: “uncertain” is a designed output of the tool, and misses are shown on purpose.

A Each gray bar is a chromosome hanging from its start; mammals and horse EquCab3 share one scale, rice, barley and potato have their own (scale bar beside each block). Box: final call with every input (mammals: FASTA, genes, Hi-C; plants: FASTA, Hi-C), colored by outcome, at least 0.75 mm tall; black bar: truth, at least 0.26 mm. Open boxes and arrows: barley's satellite calls, moved by Hi-C. Mammals: 149/157 right (4 confidently wrong, 3 uncertain wrong, 1 no call; chance 7.6; $P < 10^{-100}$, Poisson-binomial). Rice 12/12, barley 6/7, potato 2/11 (2 confidently wrong, 7 uncertain), horse EquCab3 3/5 (all uncertain); together 23/35, chance 2.3 ($P = 2.5 \times 10^{-20}$). Confident calls were right in 166 of 172 cases, uncertain or missing ones in 6 of 20 (Fisher's exact $P = 1.9 \times 10^{-12}$).

B–F Tracks, top to bottom: Hi-C score per bin (solid: anchored on the satellite calls of the other chromosomes; dotted: prior-free; 1× = chromosome median; triangles: each line's peak; hatched: bins without Hi-C data, usually the unmappable array itself); every satellite array (dark: the family cenfuse ranked centromeric; light: all others; at least 0.12 mm wide); final call; truth. Windows: calls and truth ± 3 Mb (mammals), whole chromosome (plants). B was chosen by rule: of the great-ape right calls, the one nearest the medians of call width (3.3 Mb) and call-to-truth offset (0.45 Mb) of all 149 right mammal calls. C and D are the only great-ape misses. C: two small satellite clusters (22 and 23 kb); cenfuse took the first and flagged the second, which lies under CENP-A and beside which both Hi-C peaks sit. D: CENP-A sits on 4 small arrays (84 kb in all) of families ranked centromeric, 14 Mb from the 3.4-Mb main array cluster that cenfuse called; only the anchored Hi-C peak points there, too weakly to overrule. E: the barley move of median length (lower median of 6); barley's satellite line follows a repeat family of distal chromosome regions, so the Hi-C test, anchored on those calls, fails; the arm-fold test passes and the call moves to the fold. Contact map: 836,986 cis read pairs, 4-Mb bins, balanced, observed/expected per distance, each pair of loci drawn above its midpoint, up to 300 Mb apart; in a Rabl nucleus the two arms lie side by side, so the fold appears as a vertical stripe above the centromere. F: the median-distance miss among potato's 7 uncertain misses; small arrays of the centromeric family are scattered along the chromosome, none under CENH3, and Hi-C failed its test.

G One square per chromosome, sorted by outcome; black tick: chance expectation; numbers: right/total. Mammals with FASTA only: 148/157 right (5 confidently wrong, 6 uncertain right, chance 11.1); with every input (FASTA, genes, Hi-C): 149/157 (4 confidently wrong, 3 uncertain right, chance 7.6). Pig holds 3 of the 4 confident mammal misses; its truth is CENP-A ChIP-seq from other animals, so some may be truth errors.

H Share of chromosomes right for each line alone (satellite call, prior-free Hi-C peak, anchored Hi-C peak) and for the fused call, every input (mammals: FASTA, genes, Hi-C; plants: FASTA, Hi-C); gray band: chance for the most generous line. Mammals: satellites 149, anchored Hi-C 45, prior-free Hi-C 32 of 157. Rice: satellites 12/12. Barley: prior-free Hi-C 6/7, satellites 0/7. Potato: no line above 2/11. Plants pooled: satellites 17/35, prior-free Hi-C 10/35, anchored Hi-C 10/35, fused 23/35.

I Hi-C test, for each chromosome: on each of the other chromosomes, the Hi-C peak (scored against windows at the remaining chromosomes' satellite calls) must land on that chromosome's own satellite call more often than chance, that is, on a majority of them with Poisson-binomial $P \leq 0.01$. Within a genome it passed on every tested chromosome or on none. Real Hi-C passes in 8 of 14 mammals, rice and both *Poa* species; it fails in pig, horse T2T, rhesus macaque, sheep, goat, giraffe, barley, potato and horse EquCab3, where Hi-C is then ignored (horse EquCab3 has no confident satellite call, so its uncertain calls serve as the references and only agreements at clean peaks count). Randomized Hi-C (read ends moved, or partners re-paired) passes on 0 chromosomes in 29 runs. The arm-fold test (Rabl nuclei) passes only in barley and *Poa annua*, never on randomized Hi-C. Mammals without functional truth appear only here.

Exceptions. Horse T2T gains one right call and loses one call (no call) when genes and Hi-C are added together (FASTA only to every input). Horse EquCab3 makes no confident call at all. Examples: B, E and F by the stated median rules; C and D because they are the only great-ape misses, so the typical failure cannot be chosen among more.

Analysis: asc-oulab:/data2/chris/centromere_benchmark/v2/v3 · Script: /data2/chris/centromere_benchmark/v2/v3/v4/v5/figures/benchmark_overview/benchmark_overview.py · 2026-10-27